EE 230 DESIGN: TEMPERATURE INDICATOR

Build a circuit that takes the signal from a temperature measurement sensor (e.g., the Texas Instrument LM35 from your lab kit) and produces a visual indication of certain temperature levels. In addition, the circuit will have slightly different behavior depending on if the temperature is increasing or decreasing.

Design Requirements

As the temperature is increasing:

- A green LED will be lit for temperatures below 40 °C. It will turn off when the temperature becomes greater than 40 °C.
- An orange LED will turn on when the temperature increases past 40 °C.
- A red LED will turn on when the temperature increases past 100 °C. Note that the orange LED will still be turned on.

As the temperature is decreasing:

- Once the red LED has been turned on, it will not turn off until the temperature decreases below 80 °C.
- The orange LED will not turn off until the temperature drops below 30 °C.
- At 30 °C, the green LED will once again turn on.
- You can use up to two DC power supplies to power your circuit.

Comments

- You can obtain an LM35 sensor from your lab kit. However, you don't really need the sensor in order to design the indicator circuit. You can easily mimic the electric behavior of the LM35 using a variable DC voltage source, since the output of the sensor is a DC voltage that is proportional to temperature in degrees Celsius : $v_o = (10\ mV/°C) \cdot T$, where T is the temperature in °C. However, in the end, it is fun to have an LM35 to put into the circuit as a finishing touch to your design. In using a DC source as a surrogate for the sensor, you might want to use a voltage divider circuit to divide down a larger DC voltage in order to make testing easier.
- Since the voltage from the sensor is quite small, you might want to amplify it on the way to the comparators. This will make setting the comparator switching points easier.

Testing / Reporting

- Your lab supervisor will test your circuit.
    - You should demonstrate the the LEDs turn on and off at the correct input voltages.
- You should measure the voltage at which each LED switches state. Create a table of these values and include it in your lab report.
- Write a short report that includes:
    - Complete circuit diagram
    - Photograph of your circuit
    - Written description of the design of the circuit.

- Any additional discussion about the performance of your circuit, or challenges encountered during the design.